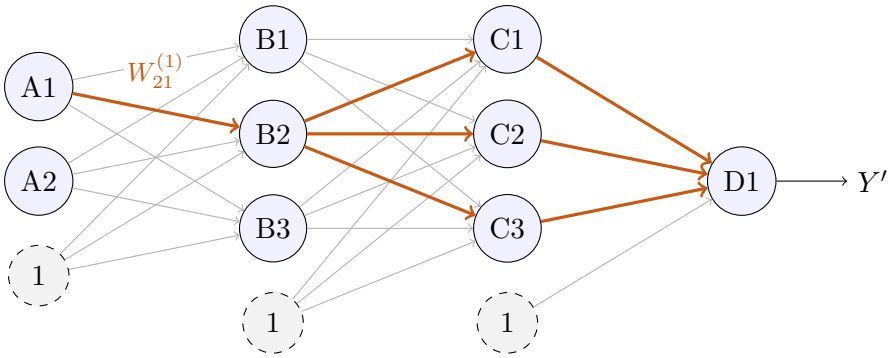
Backpropagation practice problem: solution



The thick lines are the paths along which $W_{21}^{(1)}$, the weight from A1 to B2, affects the loss: through B2, then through each of C1, C2 and C3, to D1.

Problem. The B layer uses a sigmoid, the C layer ReLU, and D1 is linear. The loss is squared error, with $\partial E/\partial Y' = Y' - Y$ (as given in the handout). A record with actual value $Y = 7$ is passed in. Using the weights and node values below, adjust $W_{21}^{(1)}$ with a learning rate of 1.

Weights (row j of each matrix holds the weights into node j of that layer):

$$\boldsymbol{W}^{(1)} = \begin{pmatrix} 2.5 & -0.4 \\ 0.2 & 2.6 \\ 1.3 & 1.5 \end{pmatrix}, \quad \boldsymbol{b}^{(1)} = \begin{pmatrix} -1.8 \\ 2.9 \\ 0.7 \end{pmatrix}, \quad \boldsymbol{W}^{(2)} = \begin{pmatrix} -1.2 & 0.7 & 2.7 \\ 0.3 & 0.5 & -0.4 \\ 1.3 & 1.9 & -3.4 \end{pmatrix}, \quad \boldsymbol{b}^{(2)} = \begin{pmatrix} 0.9 \\ 1.2 \\ 2.2 \end{pmatrix},$$

$$\boldsymbol{W}^{(3)} = \begin{pmatrix} 0.3 & 0.6 & -0.2 \end{pmatrix}, \quad b^{(3)} = 0.5.$$

Node outputs during the forward propagation (given; no forward pass needed):

A layer	B layer	C layer	D layer
$A_1 = 3$	$B_1 = -1.5$	$C_1 = 0$	$D_1 = Y' = 5.3$
$A_2 = 4$	$B_2 = 2.5$	$C_2 = 3.3$	
	$B_3 = -3.4$ (version 1), 3.4 (version 2)	$C_3 = 4$	

Notation, as in class: $W_{ji}^{(l)}$ is the weight from node i of layer $l-1$ to node j of layer l (the handout's W_{ji}^{l}). So $W_{21}^{(1)}$ goes from A1 to B2, and $W_{k2}^{(2)}$ goes from B2 to Ck. We write z for a node's input sum (its pre-activation), e.g. z_{B_2}.

The two versions of the handout differ only in how ReLU treats an input of exactly 0:

	ReLU	**ReLU slope**
Version 1	$f(x) = 0$ if $x \le 0$; $f(x) = x$ if $x > 0$	0 if $x \le 0$; 1 if $x > 0$
Version 2	$f(x) = 0$ if $x < 0$; $f(x) = x$ if $x \ge 0$	0 if $x < 0$; 1 if $x \ge 0$

(Version 2 also gives $B_3 = 3.4$ instead of -3.4, and an $A_3 = 5$ that the network does not have. Neither B_3 nor A_3 lies on a path from $W_{21}^{(1)}$ to the loss, so they do not change the answer.)

Solution

Work backwards from the loss, one layer at a time.

Step 1: the output. D1 is linear, so its slope is 1 and

$$\frac{\partial E}{\partial z_D} = \frac{\partial E}{\partial Y'} = Y' - Y = 5.3 - 7 = -1.7.$$

Step 2: the C layer. Each Ck feeds D1 through $W_{1k}^{(3)}$, and then passes the error back through its ReLU:

$$\frac{\partial E}{\partial z_{C_k}} = -1.7 \cdot W_{1k}^{(3)} \cdot \mathrm{ReLU}'(z_{C_k}).$$

C2 and C3 output positive values (3.3 and 4), so their slope is 1 in both versions:

$$\frac{\partial E}{\partial z_{C_2}} = -1.7 \cdot 0.6 \cdot 1 = -1.02, \qquad \frac{\partial E}{\partial z_{C_3}} = -1.7 \cdot (-0.2) \cdot 1 = 0.34.$$

This is where the versions differ: C1. C1 outputs 0, which happens at $z_{C_1} = 0$ or below.

- **Version 1:** an output of 0 means $z_{C_1} \le 0$, where the slope is 0. C1 passes nothing back:
$$\frac{\partial E}{\partial z_{C_1}} = -1.7 \cdot 0.3 \cdot 0 = 0.$$
- **Version 2:** here the slope at an input of exactly 0 is 1. We take C1's input sum to be exactly 0, which is what gives an output of 0 under this definition's $x \ge 0$ branch:
$$\frac{\partial E}{\partial z_{C_1}} = -1.7 \cdot 0.3 \cdot 1 = -0.51.$$
(If z_{C_1} were negative instead, the slope would be 0 and the answer would be the same as in version 1.)

Step 3: B2 adds up the three paths. B2 feeds C1, C2 and C3 through $W_{12}^{(2)} = 0.7$, $W_{22}^{(2)} = 0.5$ and $W_{32}^{(2)} = 1.9$, so

$$\frac{\partial E}{\partial B_2} = \sum_k \frac{\partial E}{\partial z_{C_k}} W_{k2}^{(2)} = \frac{\partial E}{\partial z_{C_1}} \cdot 0.7 + (-1.02) \cdot 0.5 + 0.34 \cdot 1.9.$$

Version 1: $0 - 0.51 + 0.646 = 0.136$. Version 2: $-0.357 - 0.51 + 0.646 = -0.221$.

Step 4: through B2's sigmoid. The sigmoid's slope is $B_2(1 - B_2) = 2.5 \cdot (1 - 2.5) = -3.75$,* so

$$\frac{\partial E}{\partial z_{B_2}} = \frac{\partial E}{\partial B_2} \cdot (-3.75).$$

Version 1: $0.136 \cdot (-3.75) = -0.51$. Version 2: $-0.221 \cdot (-3.75) = 0.8288$.

Step 5: the weight. $z_{B_2} = W_{21}^{(1)} A_1 + W_{22}^{(1)} A_2 + b_2^{(1)}$, so $\partial z_{B_2}/\partial W_{21}^{(1)} = A_1 = 3$:

$$\frac{\partial E}{\partial W_{21}^{(1)}} = \frac{\partial E}{\partial z_{B_2}} \cdot A_1 : \qquad \text{Version 1: } -0.51 \cdot 3 = -1.53, \qquad \text{Version 2: } 0.8288 \cdot 3 = 2.4863.$$

Step 6: the update, $W \leftarrow W - \eta\, \partial E/\partial W$ with $\eta = 1$:

Version 1: $W_{21}^{(1)} \leftarrow 0.2 - (-1.53) = \mathbf{1.73}$, Version 2: $W_{21}^{(1)} \leftarrow 0.2 - 2.4863 = \mathbf{-2.2863}$.

One node's slope, at C1, changes the sign of the update: in version 1 the path through C1 is switched off, and in version 2 it carries enough error to outweigh the other two paths.

*The node values were made up for practice. A real sigmoid always outputs a value between 0 and 1, so its slope $B(1-B)$ is always positive.